AirCast hourly PM2.5 estimation

Classical machine learning with weather and calendar features

Adrij Sarkar | B.Tech second year | CFA induction analytics

Executive summary

AirCast tests whether weather and calendar information can estimate PM2.5 concentration for a given hour in Beijing. A Random Forest selected using 2013 validation data achieves a 2014 test RMSE of **66.695 µg/m³**, MAE of **45.266 µg/m³** and R^2 of **0.491**. It reduces test RMSE by **6.38%** relative to clipped Linear Regression and by **28.69%** relative to the training-mean baseline.

The model captures useful variation, but its largest errors occur during severe pollution episodes. Hours above 250 µg/m³ make up 7.7% of the test set yet contribute 50.5% of squared error. The evidence supports weather-based estimation as a useful baseline; it does not establish reliable pollution-alert performance or a future forecasting system.

Research question and prediction setting

The prediction target is the measured concentration for the same hour as the weather inputs. Calendar variables are known in advance, but same-hour meteorological measurements are assumed available when estimating PM2.5. Consequently, this is **contemporaneous estimation**. A next-day forecast would need a separately defined horizon and only information available before that horizon.

Chronological study design

Stage	Period	Labelled hours	Role
Training	2010-2012	24,418	Fit preprocessing and models
Validation	2013	8,678	Choose model and settings
Final test	2014	8,661	Evaluate the selected approach

The three splits contain 41,757 measured targets. Final models remain fitted on 2010-2012; they are not refitted on training plus validation data. Selection uses validation RMSE, while the final test is used for reporting and error diagnosis.

Dataset and measurement scope

The UCI Beijing PM2.5 dataset contains 43,824 hourly rows and 13 raw columns from 2010-2014 [1]. PM2.5 is recorded at the US Embassy, while meteorology comes from Beijing Capital International Airport. The location mismatch means weather inputs are imperfect proxies for conditions at the pollution monitor.

Fields include a row identifier; year, month, day and hour; PM2.5 in µg/m³; temperature and dew point in °C; pressure in hPa; categorical wind direction; cumulative wind speed; and cumulative hours of snow and rain. Cumulative weather fields should not be interpreted as instantaneous measurements [1].

Evidence basis. Numerical results and figures reproduce the completed study documented in the original report [7]. Percentage improvements are calculated from its rounded metrics. This revision strengthens the explanation; it does not add a new model-training run.

Data quality and exploratory findings

Missing PM2.5 affects **2,067 rows or 4.72%** of the full table. The saved checks found no duplicate timestamps and no missing values in the selected weather/calendar inputs. Rows without targets are excluded from fitting and scoring; the PM2.5 column itself is retained. The full hourly table remains available for coverage analysis.

In training data, the mean is 97.82 µg/m³, median 73.0, standard deviation 89.25 and maximum 994. Skewness of 1.70 indicates a long upper tail. The mean exceeds the median because large events pull it upward; neither statistic alone describes the full distribution.

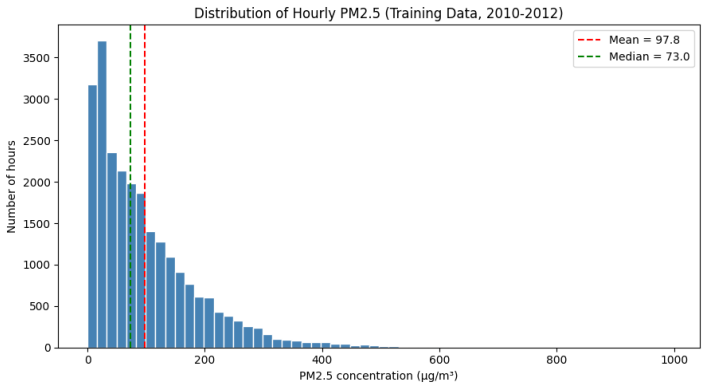


Figure 1. Training PM2.5 distribution. Many moderate readings coexist with a sparse, extended upper tail. The positive skew cautions against treating a normal-distribution rule as a calibrated event probability.

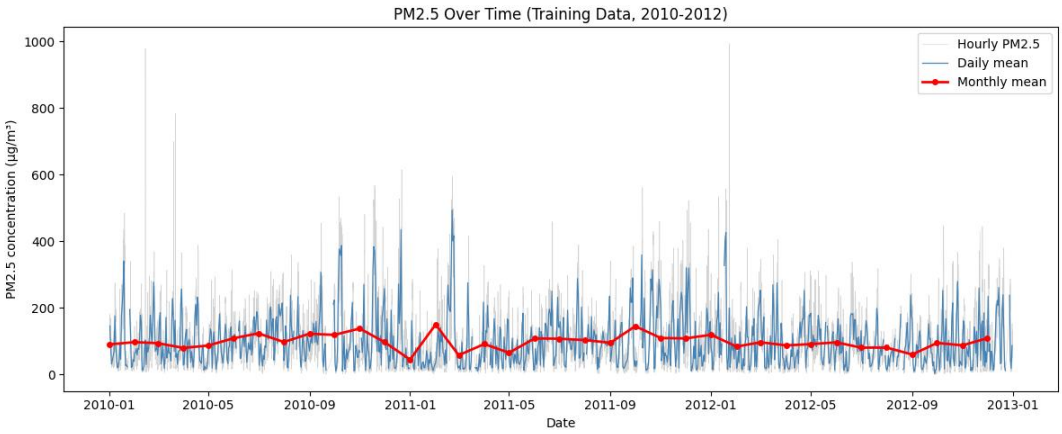


Figure 2. Training observations with daily and monthly averages. Aggregation reveals slower variation but smooths short episodes. Individual high readings remain relevant to error assessment.

Calendar-month means are highest in October (118.2 µg/m³) and lowest in May (81.1). Hourly means peak at 01:00 (113.1) and reach a minimum at 15:00 (84.9), close to 16:00 (85.0). These measured associations motivate cyclic features; they do not isolate emissions or atmospheric causes.

Outlier flags are diagnostic evidence

Training rule	Upper threshold	Flagged hours
IQR upper fence	305.5 µg/m³	797
Absolute Z-score greater than 3	Approximately 365.6 µg/m³	422

The methods disagree because IQR uses quartiles while the Z-score uses a mean and standard deviation influenced by the long tail. A flagged value is not proof of sensor error. Plausible high-pollution observations are retained so the evaluation includes the episodes the model most needs to handle.

Preprocessing features and model selection

Preserve the meaning of missing observations

Target imputation would create artificial labels, so missing PM2.5 rows do not enter supervised fitting or scoring. Training-period target missingness is 7.17% and varies by month. Excluding these rows therefore yields performance for *measured hours*; it does not establish performance during monitor outages or prove that missingness is random.

Feature construction

Engineered feature	Calculation and modelling purpose
Relative humidity	Magnus approximation using temperature and dew point, capped at 100%. Represents moisture conditions [3].
Log cumulative wind	log(1 + Iws). Compresses the right tail while retaining order and zero values.
Precipitation indicator	1 if Is > 0 or Ir > 0, otherwise 0. Encodes whether recorded snow or rain is present.
Hour cycle	sin(2π × hour / 24) and cos(2π × hour / 24). Keeps 23:00 and 00:00 adjacent in the representation.
Month cycle	sin(2π × month / 12) and cos(2π × month / 12). Represents the December-January boundary continuously.

These five feature groups produce seven engineered columns within 13 model inputs: 12 numeric and one wind-direction field, before one-hot expansion. Raw year/day and all PM2.5 values are excluded. The transformations are row-wise and do not fit parameters on future observations.

Training-only preprocessing

A Pipeline/ColumnTransformer fits a median imputer and standard scaler on numeric training inputs, and one-hot encodes wind direction. The fitted transformations are then applied to validation and test data [4]. With no missing selected numeric inputs, imputation is a safeguard rather than a source of new information. Scaling provides a common pipeline; trees do not require standardised units.

Baselines and model capacity

Model	Role and candidate settings
Training-mean baseline	Tests whether features improve on a single constant prediction.
Linear Regression	Interpretable additive baseline for comparison with nonlinear models.
Decision Tree	Depths 3, 5, 8, 12, 16 and unrestricted; selected depth 8, min_samples_leaf=20.
Random Forest	200 trees; max_features=0.5; leaf sizes 5, 20 and 50; selected min_samples_leaf=20; random_state=42.

The forest averages trees and can represent nonlinear relationships and interactions [5]. Validation RMSE selects the model because it penalises large misses. All three forest leaf settings are within 1 μg/m³ on one validation year, so the exact setting has limited ranking evidence. No statistical significance or multi-year stability is established.

Evaluation and practical interpretation

All final metrics use predictions clipped at zero: $\hat{y} = \max(0, \text{raw prediction})$. For a nonnegative target, clipping an impossible negative estimate cannot increase its absolute or squared error. Linear Regression validation RMSE falls from 73.987 to 73.430; the forest is selected under either convention. Earlier raw-output tables should not be mixed with this final comparison.

Model	Val RMSE	Test MAE	Test MSE	Test RMSE	Test R^2
Random Forest	71.889	45.266	4448.241	66.695	0.491
Linear Regression	73.430	49.511	5075.071	71.240	0.420
Decision Tree	76.833	50.077	5395.474	73.454	0.383
Training mean	98.142	68.893	8747.145	93.526	-0.000

MAE and RMSE are in μg/m³; MSE is in (μg/m³)²; R^2 is dimensionless. Values reproduce the rounded original results. The baseline R^2 is slightly negative and rounds to -0.000.

What the improvement means

Relative to Linear Regression, forest test RMSE falls by 4.545 μg/m³ (6.38%) and MAE by 4.245 μg/m³ (8.57%). Relative to the mean baseline, test RMSE falls by 26.831 μg/m³ (28.69%). These gains show predictive value beyond a constant or linear estimate, but the remaining MAE of 45.266 μg/m³ is substantial.

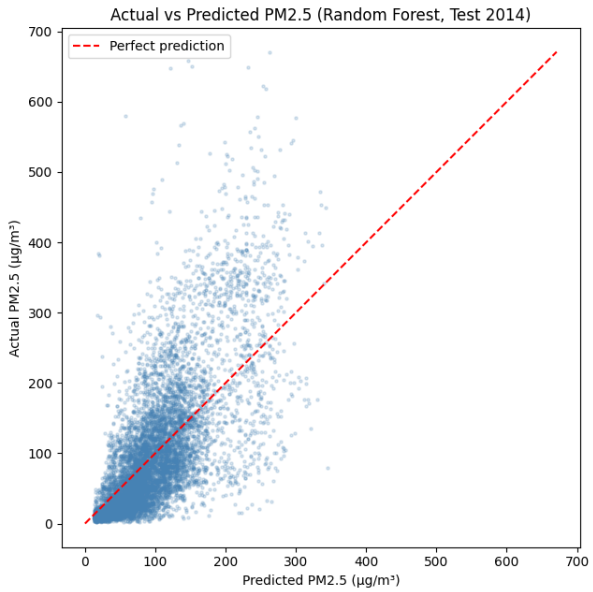


Figure 3. Actual versus predicted PM2.5 in 2014. Actual concentration is on the vertical axis; points above the diagonal are underpredictions. High observed concentrations are frequently compressed into lower predictions.

With residual e = actual − predicted, MAE = mean(|e|), MSE = mean(e^2), and RMSE = √MSE. R^2 = 1 − SSE/SST, with SST measured around the evaluated split's own mean [6]. Thus R^2 = 0.491 is a reduction in squared error relative to that reference, **not 49.1% accuracy**.

Forest training R^2 is 0.653, versus 0.463 on validation and 0.491 on test. This gap indicates weaker performance beyond training. A higher test R^2 than validation R^2 can reflect different years and target variability; it does not show that overfitting has disappeared.

Error concentration and feature reliance

Residuals are defined as actual minus predicted, so positive values mean underprediction. The overall mean residual is only +1.61 µg/m³, but this average hides offsetting errors. Residual standard deviation rises from approximately 31 to 128 µg/m³ across predicted-concentration groups.

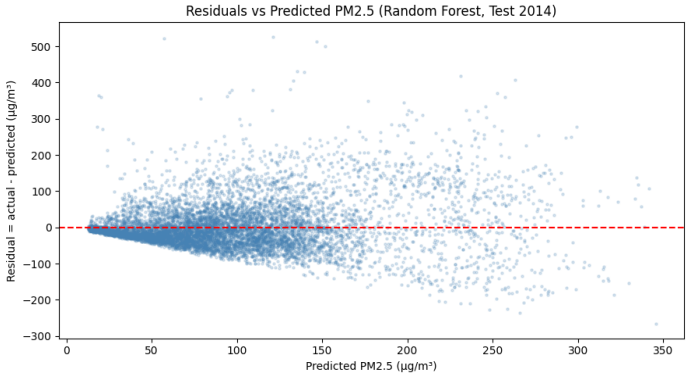


Figure 4. Residual spread increases with predicted concentration. A single average error does not describe uncertainty equally well across pollution levels.

Severe events dominate squared error. Test hours above 250 µg/m³ contribute 50.5% of squared error despite representing 7.7% of observations. Based on these rounded shares, mean squared error per hour is about **12.2 times** larger in this group than in the remaining hours. This threshold is an analytical grouping, not a health-standard classification.

February has the highest monthly RMSE (113.4 µg/m³); June is overpredicted on average (mean residual −36.7). The maximum test prediction is 345.6 against a maximum observed value of 671.0 µg/m³. These maxima need not occur at the same hour, and 345.6 is not a universal model ceiling.

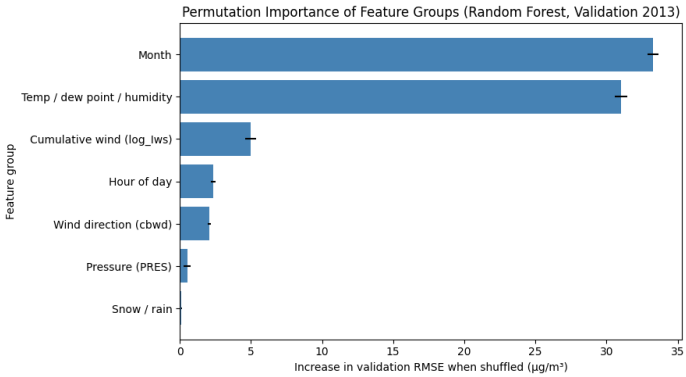


Figure 5. Grouped validation permutation importance. Bars show mean RMSE increase across five shuffles; error bars show shuffle standard deviation, not confidence intervals across years.

Feature group removed and model refitted	Validation RMSE increase
Temperature, dew point and humidity	13.572 µg/m³
Month features	4.183 µg/m³
Wind inputs	1.844 µg/m³

Permutation measures how much a fitted model relies on information when it is disrupted; ablation asks how well a refitted model performs without that information. Correlated predictors can substitute for one another and shuffling can create unusual combinations [8]. Different rankings are therefore possible. Both analyses describe predictive reliance, not causal effects of weather or season.

Conclusions limitations and next experiments

Conclusion. Random Forest is the strongest tested estimator under the stated split and clipping convention. It improves aggregate error, but rare high-concentration episodes remain the central weakness. The next objective should be reliable evaluation of those failures, not a claim of operational forecasting readiness.

Limits of the evidence

Coverage and transfer. Uneven missing labels and separated monitoring locations limit representativeness. A single city and historical period do not demonstrate transfer to other stations or current conditions. **Available information.** Same-hour weather omits emissions and recent pollution history. Tree averaging is consistent with smoothed extremes, but the present diagnostics do not establish the unique cause of each error.

Evaluation uncertainty. One validation year supports only a limited comparison. Hourly observations can be dependent, so the number of rows is not the number of independent episodes. Confidence intervals and expanding-window comparisons are not reported. Because 2014 outcomes are now known, subsequent model changes need a new untouched evaluation period.

Prioritised next experiments

1. Reproduce and test stability. Run the notebook from a clean kernel, pin packages and data, and compare models on identical expanding-window folds. Report fold-to-fold RMSE and severe-event errors.
2. Add temporal information responsibly. Build historical features on the complete hourly grid. For future prediction, define a horizon, shift PM2.5 lags and rolling statistics, and exclude future observed weather.
3. Evaluate the intended decision. Compare high-pollution MAE and underprediction rates alongside overall RMSE. Any event-weighted loss or alert rule is a proposed experiment and must be selected without the final test.

Reproducibility and assistance

The original study documents a corrected notebook, result CSVs, recovered plots, dependencies and reproduction notes [7]. Original package versions and a frozen raw-data snapshot were not supplied in that study; a clean-kernel rerun remains to be confirmed. Gemini assistance with pipeline/diagnostic design and formatting, and ChatGPT assistance with code, interpretation and documentation, were disclosed. This revision adds editorial and arithmetic checks; the author remains responsible for verifying and explaining the work.

References and evidence

[1] Chen, S. (2015). Beijing PM2.5. UCI Machine Learning Repository. DOI 10.24432/C5JS49. CC BY 4.0. Source link.

[2] Liang et al. (2015). Assessing Beijing's PM2.5 pollution: severity, weather impact, APEC and winter heating. Proc. R. Soc. A 471, 20150257. Source link.

[3] Alduchov and Eskridge (1996). Improved Magnus Form Approximation of Saturation Vapor Pressure. Journal of Applied Meteorology 35, 601-609. Source link.

[4] Scikit-learn. Common pitfalls and recommended practices. Source link.

[5] Scikit-learn. RandomForestRegressor API. Source link.

[6] Scikit-learn. Metrics and scoring, regression metrics. Source link.

[7] Scikit-learn. Permutation feature importance. Documentation links checked 27 October 2026. Source link.